Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ____________
Topic: Applications of Systems (Word Problems) **Date:** ________

Foundation (Jalapeno)

Open Ended Questions (FRQ)

Q9. Tom and Alex are on a biking trip. Tom travels x miles per hour and Alex travels y miles per hour. If they travel for t hours and Tom travels 5 miles more than Alex, write a system of equations to represent this situation and find the values of x and y if $t = 2$ hours and Tom travels a total of 26 miles.

Q10. A company is manufacturing basketballs and soccer balls. They produce x basketballs and y soccer balls per day. If the total number of balls produced per day is 50 and $x = 2y - 5$, write a system of equations to represent this situation and find the values of x and y if the company produces 15 soccer balls per day.

Chef's Tips

- Emphasize the importance of reading the problem carefully and identifying the key elements of the system.
- Encourage students to use a variety of methods to solve the system, such as substitution or elimination.
- Remind students to check their solutions by plugging them back into the original equations.